

DR. OGECHI IGBOANUGO (CCIE-EI # 50094, JNCIE-SP #3027, JNCIE-ENT #611)

Wylie, Texas

Phone: 501-612-8191; **Email:** oigboanugo_cisco@yahoo.com

SUMMARY

Doctor of Information Technology, and Skilled CCIE & 2 X JNCIE Sr. Staff Network Architect & Engineer, in local and wide area networks (LAN/WAN/DC/Campus) who offers technical services and has recognized strengths in problem-solving, troubleshooting, planning/implementing proactive procedures to avoid problems in the network. Looking for a position as a Network Engineer/Architect to implement effective installation, configuration and maintenance of IT network systems.

WORK EXPERIENCE (OVER 14 YEARS)

FRONTIER COMMUNICATIONS USA

Sr Staff Network Architect/Director, Network Design Team (6/2/2020 to Present)

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- Subject Matter Expert providing guidance to Senior Engineers/Architects in Frontier's Metro Ethernet networks working with Cisco, Juniper & Nokia network environments;
 - Manager of the Network Design/Network Planning Team for Video, Metro E & Access Networks;
 - Deployed, troubleshoot and managed Leaf-Spine data center architectures for Cisco and Arista EOS L2LS & L3LS architectures.
 - Proficient in Arista Validated Design (AVD) for DevOps automation of network deployment.
 - Proficient in Cloud Vision Portal (CVP) for Network management and configuration.
 - Proficient in Nokia NSP for Network management and configuration of Nokia Devices.
 - Deployed an entire 7750 SR Network from scratch leveraging protocols like Segment Routing, IS-IS, MPLS, BGP and so on;
 - Provided connectivity to AWS Cloud services via AWS Direct Connect over dedicated circuits between AWS and my Video data center, thereby connecting the Video Data center(s) to desired VPCs; BGP peering was used.
 - Achieved AWS site-to-site VPN connectivity by connecting disparate Video networks to AWS VPCs via IPSEC tunnel establishment over an internet connection;
 - Firewall administration through access-control & policy configuration on Juniper SRXs, Cisco ASAs, etc
 - Review MPLS Networks and determine network vulnerabilities;
 - Ensure Network appliances have up-to-date software OSs;

- Conducted network-wide audits and created multiple network diagrams across various networks;
- Managed the total health of the Legacy Metro E Network made up of PE routers which are Nokia 7210 routers and PE/P routers are Nokia 7450s, and a separate MPLS Network comprised of Cisco ASR 9000s and another Metro E Juniper MPLS network made up of MX960/MX480;
- Troubleshooting of 7750 Metro E based networks that provide EIA and Business VPN services to customers; have configured EVPN, ELINE, VPRN, VPLS, Hairpin ELINE etc;
- Administered SAM 5620 to manage and discover Legacy (over 200 Nokia 7210s and 100 7450s) and new 7750 Metro E networks;
- Subject Matter Expert providing guidance to Senior Engineers/Architects in Frontier's FiOS Video Networks;
- Involved in outage escalations especially if an outage has been ongoing and there hasn't been a resolution yet; Tier 4 escalation point for Network Operations Teams.
- Lab testing of new features, software and documentation of network standards;
- Design and Publish Architectural documents to assist the Architecture team with standards for the network;
- **Capacity planning** and circuit migrations for Metro E & Video networks, ensuring successful migration of traffic to new circuits and to new devices;
- Managed Capacity of Metro E Networks and reduced link capacity from 99% utilization down to 60% utilization on average for two years running.
- **Migrations:** Capable of successfully implementing translation/migration of configurations from Juniper to Cisco, Cisco to Juniper, Juniper to Nokia, Nokia to Juniper and Cisco to Nokia, Nokia to Cisco routers, Cisco to Arista, Juniper to Arista;
- On a daily basis, I work with ASA & Juniper SRX firewalls, A10 & ACE load-balancers; VPN technologies (IPSEC/GRE); MPLS in the Metro E networks; BGP & OSPF; Build and maintain Lab Datacenter networks from scratch; Capacity management of network uplinks; Provide cheap and scalable solutions to daily network needs;
- **Network Automation:**

 - Design Network Automation Schema using DevOps/NetDevOps.

 - Ansible playbooks, JINJA and Python to develop network builds and push incremental updates in the network using CICD pipelines.
 - Used Python and YAML to automate the creation of device configuration for multiple devices.

FRONTIER COMMUNICATIONS USA (4/1/2017 to 6/2/2020)

Network Architect

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- Subject Matter Expert providing guidance to Senior Engineers in Frontier's Metro Ethernet networks working with Cisco, Juniper & Nokia network environments;
 - Subject Matter Expert providing guidance to Senior Engineers in Frontier's FiOS Video Networks;
 - Deployed an entire Nokia 7750 SR Network from scratch leveraging protocols like Segment Routing, IS-IS, MPLS, BGP and so on;
 - Troubleshooting of Nokia 7750 Metro E based networks that provide EIA and Business VPN services to customers; have configured EVPN, ELINE, VPRN, VPLS, Hairpin ELINE etc;
 - Administered SAM 5620 to manage and discover Legacy (over 200 Nokia 7210s and 100 7450s) and new 7750 Metro E networks;
 - Managed the total health of the Legacy Metro E Network made up of PE routers which are Nokia 7210 routers and PE/P routers are Nokia 7450s.
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- On a daily basis, I work with ASA & Juniper SRX firewalls, A10 & ACE load-balancers; VPN technologies (IPSEC/GRE); MPLS in the Metro E networks; BGP & OSPF; Build and maintain Lab Datacenter networks from scratch; Capacity management of network uplinks; Provide cheap and scalable solutions to daily network needs;
 - Involved in outage escalations especially if an outage has been ongoing and there hasn't been a resolution yet;
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FRONTIER COMMUNICATIONS USA (12/14/2015 to 3/30/2017)

Senior Lead Video Integration Engineer

- Provide Video engineering and network architecture services to the Frontier FiOS video technology division;
- One of Lead Engineers that built the Video Head-end (VHE) leveraging Data-center devices and applications; e.g Nexus platform;
- Involved in Verizon acquisition cut-over as one of Technical integration Engineers offering Tier IV services and escalation point;
- Multicast content traffic engineering and planning;
- Tier 3 support for Video Engineering, supporting the Super Hub Office (SHO), Super Head End (SHE) and VHOs (Video Hub Office);
- Responsible for SHE redesign, architecture and build;

FRONTIER COMMUNICATIONS USA (03/29/2015 to 12/14/2015)

Senior Network Engineer

- Responsible for managing network element growth, expansion and upgrades; collaborate with Network Architects on requirements and stay educated in all aspects of network standards;
- Provide Tier 4 technical support for OPT-E-MAN and IPAG Network;
- Create network element command-level configurations and Method of Procedure (MOP).
- Deployed Nexus 7009 for Video Lab, leveraging Dual VDCs, CFS, FCoE, VPC, OTV & Network Load Balancing (NLB) for VOD content servers, etc.
- Currently Nexus 7009 Tier 3 support for Video Lab;
- Utilize capacity and traffic monitoring tools to evaluate network performance, plan remediation plans, specifications;
- Maintain network drawings and documents with current configurations and supporting data;
- Apply knowledge of Cisco & Juniper network equipment, Cisco ISR routers, Nexus, Cisco Catalyst Switches, Cisco Wireless Access points, Juniper MX960s & MX480s.
- Apply knowledge of network troubleshooting, organization, network planning, maintenance, monitoring protocols and software.

Tekmark Global Solutions (09/15/2014 to 03/29/2015)

Project: FRONTIER COMMUNICATIONS USA

Senior Network Engineer

- Responsible for managing network element growth, expansion and upgrades; collaborate with network standard group on requirements and stay educated in all aspects of network standards;
- Create network element command-level configurations and Method of Procedure (MOP).

- Utilize capacity and traffic monitoring tools to evaluate network performance, plan remediation plans, specifications and create Engineering Work Request (EWR);
- Maintain network drawings and documents with current configurations and supporting data;
- Apply knowledge of networking switches, routers, IP/MPLS, QoS, BGP, OSPF, PPP, Ethernet, WiFi, Multicasting;
- Apply knowledge of Cisco & Juniper network equipment, Cisco ISR routers, Cisco Catalyst Switches, Cisco Wireless Access points, Juniper MX960s & MX480s.
- Apply knowledge of network troubleshooting, organization, network planning, maintenance, monitoring protocols and software.

TEKSystems (09/03/2013 to 09/12/2014)

Project: Windstream Communications USA

Network/ Routing Engineer Contractor III (IP Access Engineer)

- Sole **Mop** designer (method of procedure): **I managed the entire layer 3 (network) implementations for Broadband Engineering team** that creates new GE uplinks/circuits from Broadband remote Access Routers (BRAS) & BNG to various Provider & Provider Edge Routers (Juniper Mx-960s, Mx480s, m-120s,m10i) in order to increase network performance.
- **Single-handedly** managed Layer 3 network Capacity and planning for Windstream BRAS/BNG.
- Troubleshooting IP Core routers, routing protocols, and transport layer problems.
- Handled projects on network upgrades & configuration (e.g NPEG2 upgrades for cisco, ge-card upgrades for Juniper & Erickson Redback devices).
- Review of MOPS for all IP Engineering.
- Implementation of all BRAS/BNG capacity management MOPs during maintenance periods.
- Built several **MPLS** Traffic engineering LSPs for Windstream Core Networks.
- Cisco 10k PRE 2 to PRE 4 upgrade (Processor Routing Engine).
- Circuit implementations varying from Frame relay, ATM, OSPF, ISIS, BGP.
- Vendor management and liaison.
- Experience using MSS, JobTrac, E-health & Remedy ticketing.

Prometric (Test development & Delivery specialist)

Subject Matter Expert (SME for Routing & Switching) (09/07/2013 to 03/11/2014)

- Provide expertise on networking routing protocols (OSPF, EIGRP, and BGP) to review a very popular Network Routing & Switching exam and provide advice.
 - Provide expertise on switching technologies (like VLANs, LAGs, Trunking, spanning-tree, etc) to review computer network exams and provide advice.
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Arkansas Electric Cooperatives Corporation

Network Analyst Intern (05/18/2013 to 08/30/2013)

- Providing end user support and resolving all network related problems (like user-connectivity issues, capacity issues and unexpected outages).
 - Testing network performance and Management including Pro-active monitoring of LAN and WAN links as well as capacity planning. This included over 5000 nodes.
 - Circuit implementations of Ethernet & Frame relay.
 - Vendor management and liaison.
 - Ensure that all monitored sites (branch offices) are in 'up' state and have high availability.
 - Create tickets whenever a Network node goes 'down' (not –operational).
 - Design a network performance monitoring system for two of our sites, Fitzhugh and Magnet Cove sub-stations.
 - Perform Business Intelligence using SQL server 2012 to mine and analyze network data and produce relevant reports;
 - Report to various sites to solve network issues
 - IOS Upgrades on routers and switches.
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Arkansas Tech University

Graduate Assistant Network Lab Instructor (08/30/2012 to 05/03/2013)

- Taught computer Networks to over 50 US undergraduate students.
- Design of Technical Lab manual titled 'Network Analysis, a practical approach' Instructor and student copy. (120 pages each).

Each lab exercise focuses on most Network technologies contained in CCNA curriculum. Technologies like IP subnetting, access-lists, port-security, routing protocols (OSPF, RIP, IGRP) just to mention a few.
- Design and set-up Network Lab with over 20 Routers (1800 & 2800 series), and 44 switches (3550 & 2960 series).

A-Z Petroleum Services Nigeria Limited

Network Administrator (01/12/2009-12/22/2011)

- Diagnose hardware (faulty printers, monitors & CPUs) and software problems (Windows operating system issues for windows XP, Vista & windows 7), and replace defective components.
- Perform data backups and disaster recovery operations.
- Develop/implement immediate solutions for common server and LAN/WAN problems: low disk space, user permissions, file restores, TCP/IP address and/or Name conflicts, Troubleshooting LAN/WAN Backbone connectivity and trouble reporting, User Lockout, Server Software/Hardware failures, etc.
- Cable management, racking, stacking and layer one connectivity establishment
- Proactive monitoring of all network installations and device.
- Various Printer installations and management (including Ink inventory management).
- HP desktop computer installations and support.

EDUCATION:

- **Doctorate in Information Technology**, Capella University (2024)
- **Masters of science** in Information Technology, Arkansas Tech University, GPA: **3.83**.
- **Bachelors in Engineering**, Electrical Engineering, Nnamdi Azikiwe University, Nigeria.

CERTIFICATIONS:

ACE L7 # AC:E6:23:AA:00:77 Arista Cloud Architect Expert	2024
ACE L6 # AC:E6:23:AA:00:77 Arista Cloud Design Expert	2024
JNCIE-SP #3027 Juniper Networks Certified Expert - Service Provider	2021
JNCIE-ENT #611 Juniper Networks Certified Expert Enterprise	2017
CCIE Enterprise Infrastructure Expert (R&S) #50094	2015
Juniper Networks Certified Professional –SP (JNCIP-SP)	2017
Juniper Networks Certified Specialist –SP (JNCIS-SP)	2015
Juniper Networks Certified Specialist –ENT (JNCIS-ENT)	2016
Arista ACE L5 (Automation)	2023
Nokia NRS 2 [Currently working on it]	

CCIE Routing & Switching WRITTEN Vr 5 (PASSED)	2014
Cisco Certified Network Professional (CCNP) Enterprise Infrastructure	2013
Cisco Certified Network Associate (CCNA)	2013
Juniper Networks Certified Associate (JNCIA-Junos)	2013
Solarwinds Certified Professional (SCP) Network management & monitoring	2013
Cisco Certified Entry-level Technician (CCENT)	2012

REFERENCES:

Mr. Ron Robison (Dept Chair Comp & Info Sci, Associate Professor)
Arkansas Tech University, Computer and Information Science
Contact: (479) 968-0664
Email: rrobison@atu.edu

Sanford Mcallum (Network Security Analyst II)
SouthWest Power Pool, Little Rock, Arkansas, U.S.A
Contact: +1-(501)-352-3574
Email:

Daniel Hardy (**Senior** Network Analyst)
At Home Group Inc Dallas, Texas U.S.A
Contact: +1-(501)-412-6989; Email: